

Ai Impact on Students

1. Team Members

- Member 1: Cyrus Troy Bazar

2. Dataset Collection and Description

AI Impact on Students

Source: Kaggle / Public Dataset

Dataset Link: <https://www.kaggle.com/datasets/dspritom/ai-impact-on-students>

Number of Variables: total 15 columns

Number of Observations: 28856 |

Variables

Variable	Data Type	Description
Student_ID	Integer	A unique identification number assigned to each student.
Major_Category	Categorical	The student's academic major or field of study.
Year_of_Study	Categorical	The student's current year level in their academic program.
Pre_Semester_GPA	Numerical (Float)	The student's GPA before the semester began.
Weekly_GenAI_Hours	Numerical (Float)	The number of hours per week the student spends using generative AI tools.
Primary_Use_Case	Categorical	The student's main purpose for using generative AI, such as studying, coding, research, or writing.
Prompt_Engineering_Skill	Categorical	The student's level of skill in creating and designing effective prompts for generative AI tools.
Tool_Diversity	Numerical (Integer)	The number of different generative AI tools used by the student.
Paid_Subscription	Categorical	Indicates whether the student uses a paid subscription to a generative AI service.
Traditional_Study_Hours	Numerical (Float)	The number of hours per week the student spends studying without generative AI tools.
Perceived_AI_Dependency	Numerical (Integer)	A score representing how dependent the student feels on generative AI for their academic activities.
Institutional_Policy	Categorical	Indicates the level or type of institutional policy governing the use of generative AI in the student's academic environment.
Anxiety_Level_During_Exams	Numerical (Integer)	A score representing the student's level of anxiety experienced during examinations.
Post_Semester_GPA	Numerical (Float)	The student's GPA after completing the semester.

Variable	Data Type	Description
Burnout_Risk_Level	Categorical	The student's categorized level of risk for experiencing academic burnout.

Unit of Observation

Each row represents one student.

Data Collection

This data set comes with a codebook on the Kaggle site.

3. Broad Research Question

- Does Ai Impacts students performance on academics?
- Do students with high AI usage decrease their traditional study hours proportionately, and how does this tradeoff impact performance?
- Primary Use Case Impact: How do different primary uses of AI (e.g., coding assistance, essay writing, summarizing notes, or test prep) vary in their effect on student GPA?

4. Data Exploration

The dataset was first inspected to determine its size, variables, data types, missing values, duplicate records, and possible invalid values.

The dataset contains 15 variables and 28856 | observations.

Initial inspection showed that 8 variables were numeric and suitable for statistical analysis.

5. Data Cleaning and Pre-processing

The following preprocessing steps were performed:

- Checked the dataset for missing values.
- Checked for duplicate records.
- Checked whether numerical variables contained invalid values.
- Verified the data types of each variable.
- Examined extreme values and possible outliers.
- Removed or corrected invalid records where necessary.

After preprocessing, the dataset was considered suitable for exploratory analysis.

6. Missing Data Analysis

```
import pandas

df = pandas.read_csv("./data.csv")

df.info
```

Column	Non-Null Count	Dtype
0 Student_ID	28856	non-null int64

Column	Non-Null Count	Dtype
1 Major_Category	28856	non-null object
2 Year_of_Study	28856	non-null object
3 Pre_Semester_GPA	28856	non-null float64
4 Weekly_GenAI_Hours	28856	non-null float64
5 Primary_Use_Case	28856	non-null object
6 Prompt_Engineering_Skill	28856	non-null object
7 Tool_Diversity	28856	non-null int64
8 Paid_Subscription	28856	non-null object
9 Traditional_Study_Hours	28856	non-null float64
10 Perceived_AI_Dependency	28856	non-null int64
11 Institutional_Policy	28856	non-null object
12 Anxiety_Level_During_Exams	28856	non-null int64
13 Post_Semester_GPA	28856	non-null float64
14 Burnout_Risk_Level	28856	non-null object

This shows that theres no missing values in our dataset.

7. Descriptive Statistics

Demographic and AI Usage:

Metric	Student_ID	Pre_Semester_GPA	Weekly_GenAI_Hours	Tool_Diversity
count	28856.00000	28856.000000	28856.000000	28856.000000
mean	114428.50000	3.151702	9.218165	2.800353
std	8330.15402	0.478808	9.300575	1.189669
min	100001.00000	1.183000	0.000000	1.000000
25%	107214.75000	2.838000	2.350000	2.000000
50%	114428.50000	3.218000	5.950000	3.000000
75%	121642.25000	3.527000	12.980000	4.000000
max	128856.00000	3.997000	40.000000	5.000000

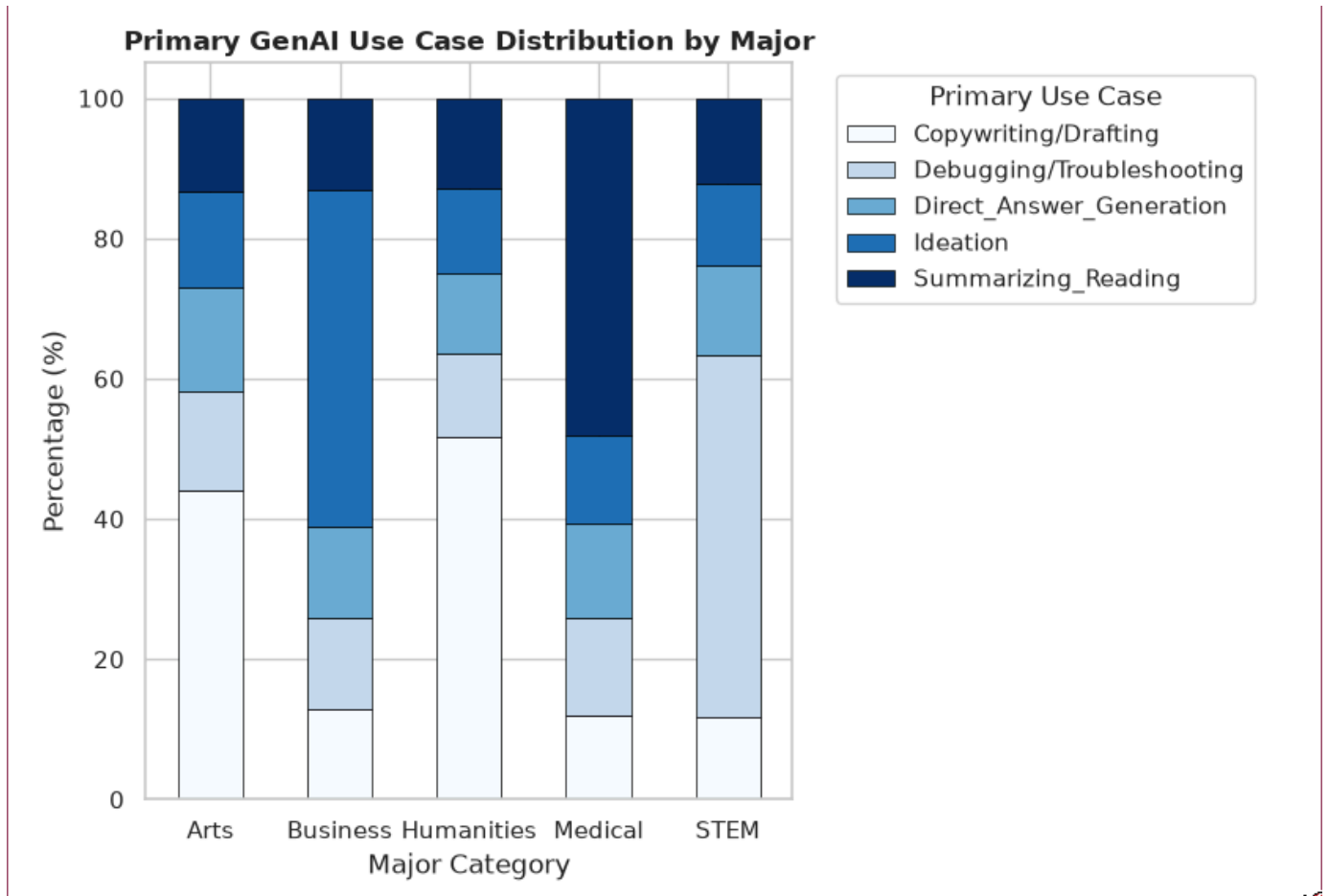
Study Habits, Wellbeing & Academic Outcomes:

Metric	Traditional_Study_Hours	Perceived_AI_Dependency	Anxiety_Level_During_Exams	Post_Semester_GPA
count	28856.000000	28856.000000	28856.000000	28856.000000
mean	11.150396	3.608296	4.344261	3.350033
std	5.187562	1.947710	2.195916	0.497251
min	1.000000	1.000000	1.000000	1.000000
25%	7.450000	2.000000	3.000000	3.025000

Metric	Traditional_Study_Hours	Perceived_AI_Dependency	Anxiety_Level_During_Exams	Post_Semester_GPA
50%	11.110000	3.000000	4.000000	3.423000
75%	14.670000	5.000000	6.000000	3.750000
max	35.860000	10.000000	10.000000	4.000000

8. Visualization

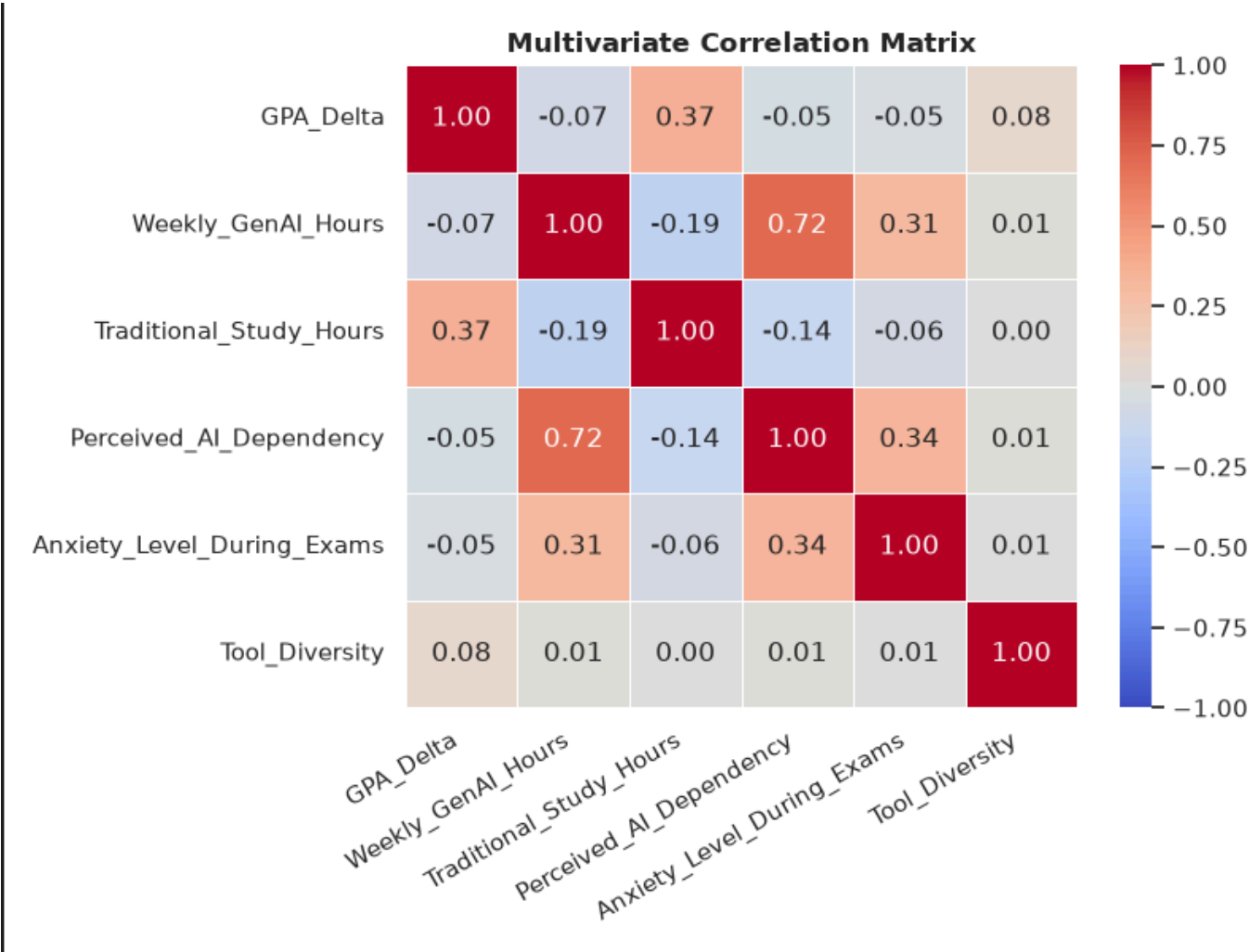
GenAI Use Case Distribution by Major 100% stacked bar:



This stacked bar shows the percentage of primary use case of ai in different Major categories.

- **Humanities** and **Arts** Students uses Ai for drafting.
- **Buseness** Students use it mostly for ideas.
- **Medical** Students use it to Summarize reading
- And **STEM** Students use it for Direct answer generations.

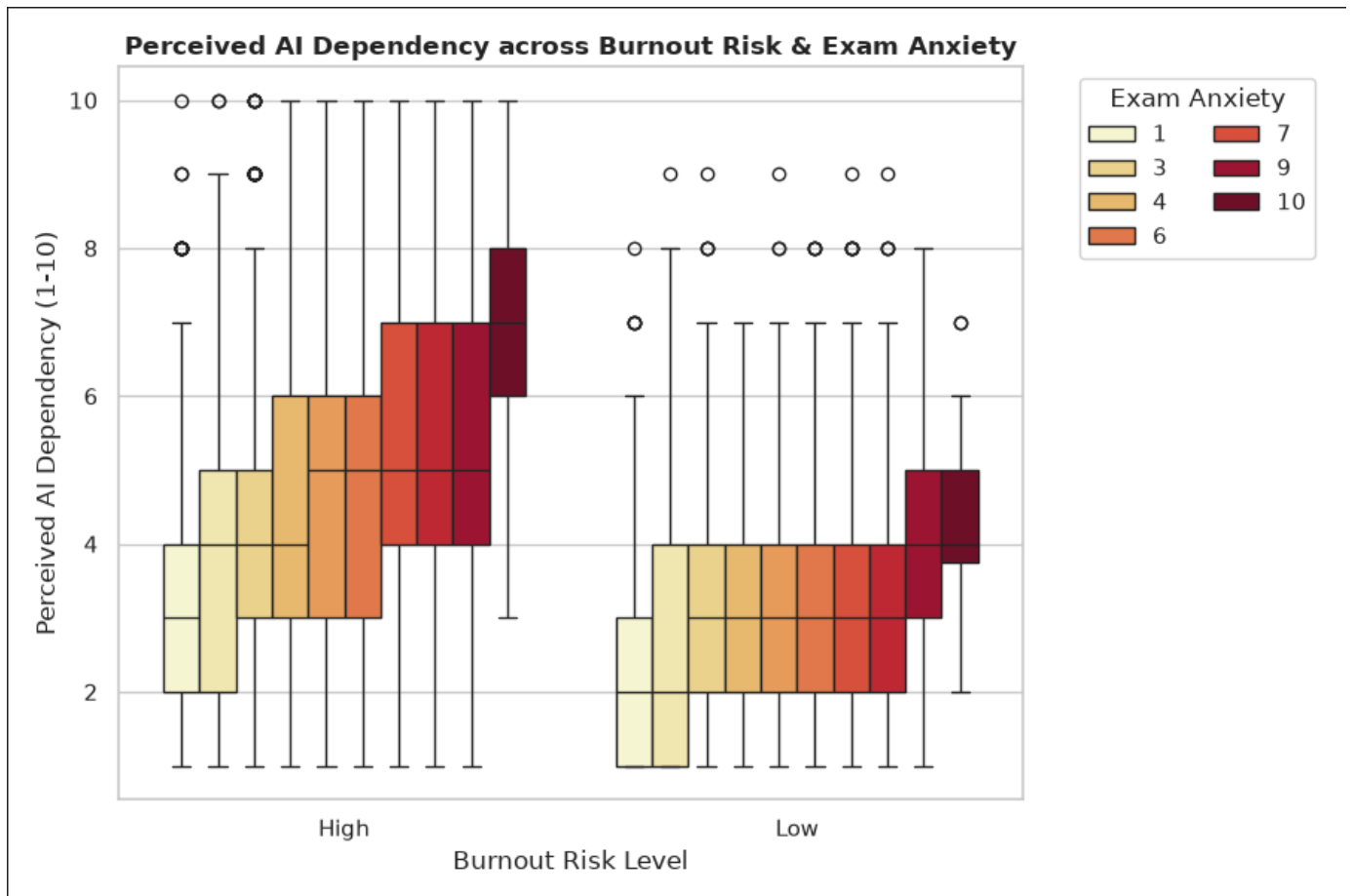
Multivariate Corelation Martix:



The correlation matrix shows the strength and direction of the relationships between GPA Delta, GenAI usage, traditional study hours, perceived AI dependency, exam anxiety, and tool diversity.

- GPA Delta & Traditional Study Hours have **0.37** Moderate positive relationship. Students who spend time on Traditional Study hours are likely to have higher GPA.
- Weekly GenAI Hours & Anxiety During Exams have **0.31** Weak to moderate positive relationship, meaning Greater Genai Usage is somewhat associated with anxiety levels during exam

Grouped Box Plot (AI Dependency vs. Burnout Risk):



The box plot shows how students' perceived AI dependency (1–10) varies according to their burnout risk level and exam anxiety level. Each box/bar represent Exam Anxiety levels.

- **Low burnout risk level:** AI dependency is generally lower, with most median values around 2–4.
- **High Burnout Risk:** Students generally show higher AI dependency with median values mostly around 3–6 depending on exam anxiety.
- **Outliers:** This also shows the data outliers as a dot.
- **Exam Anxiety:** Within the High burnout group, higher anxiety levels tend to be associated with higher AI dependency.

9. Specific Research Question

- **Specific Hypothesis:** We hypothesized that higher Ai usage on academics leads to less burnout risks and stress thinking.
- **Findings:** The graphs reveal that students spending more time on Ai most likely to develop burnout and stress.